

Exploratory (Spatial) Data Analysis

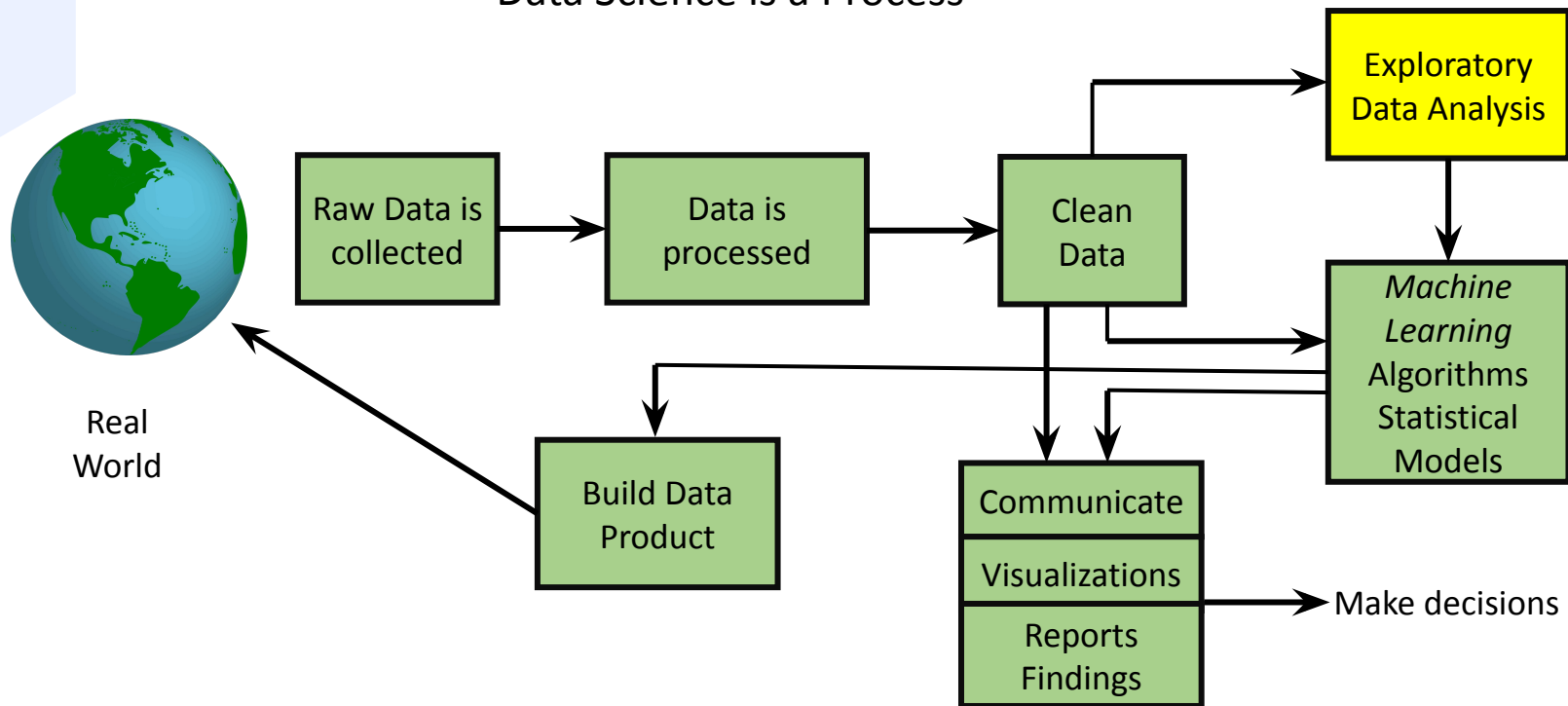
From attributes to spatial structure

Édipo Cremon
Rennan Marujo



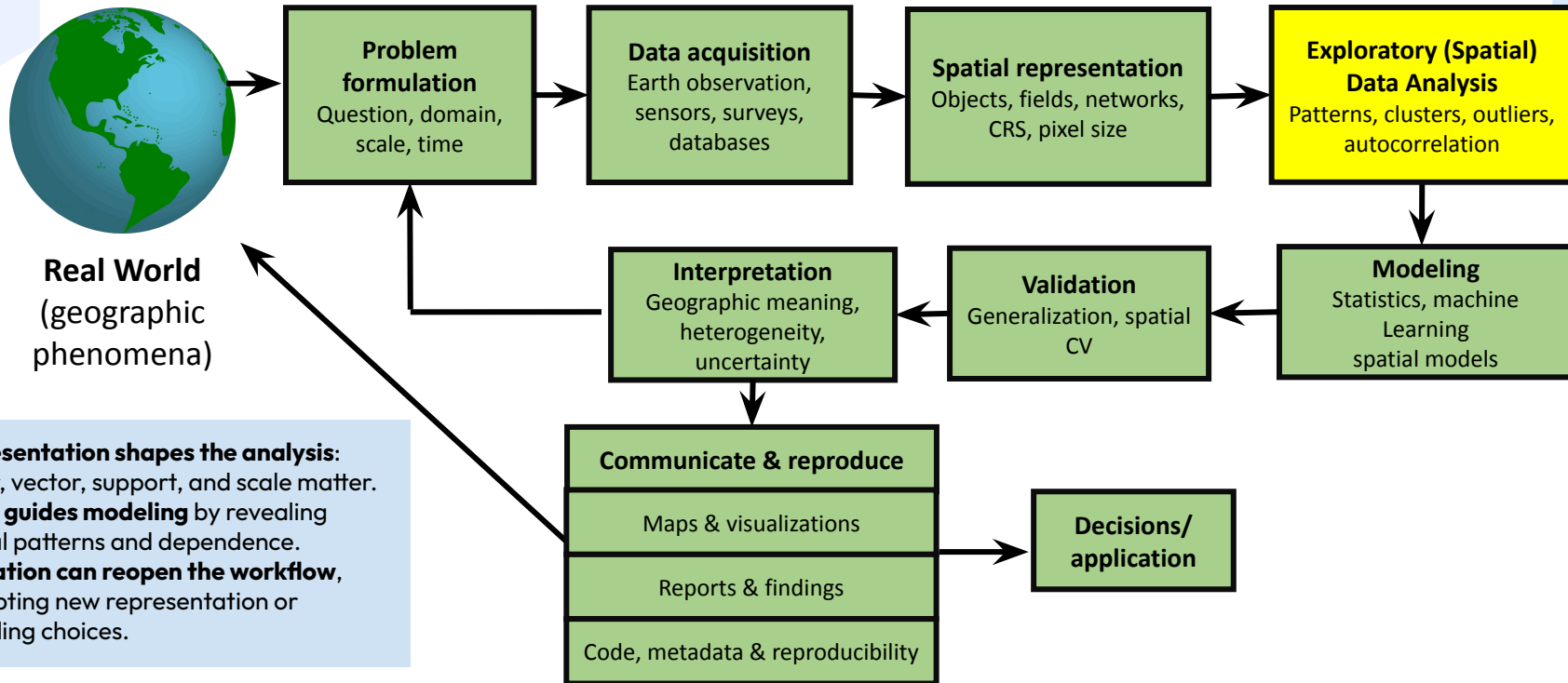
What is Data Science?

Data Science is a Process



What is Geospatial Data Science?

Geospatial Data Science is an iterative process



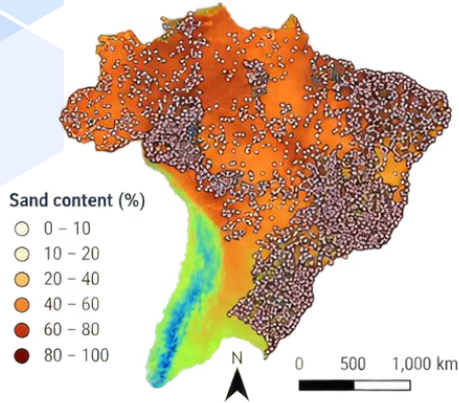
Iterative, not strictly linear.



Before modeling:
know your
data

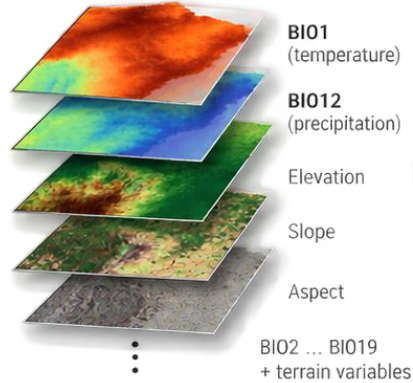
Example

Regression: Sand content (%)



Predictor variables

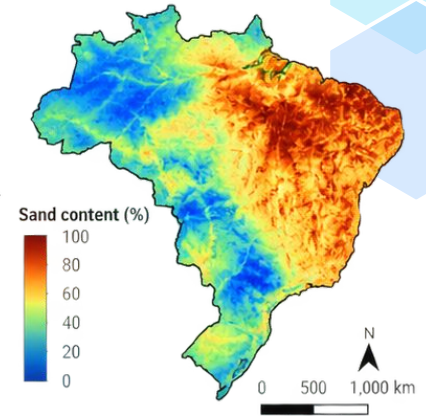
BIO1 - BIO19 + terrain variables



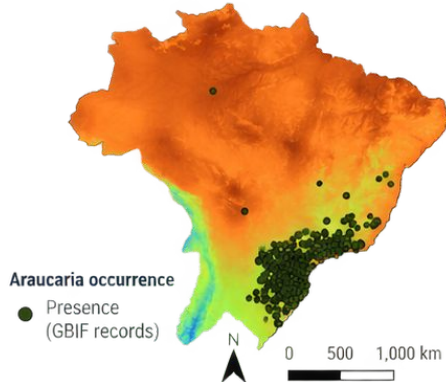
Machine learning regression model

Learn the relationship between environmental predictors and sand content (%)

Predicted sand content (%)

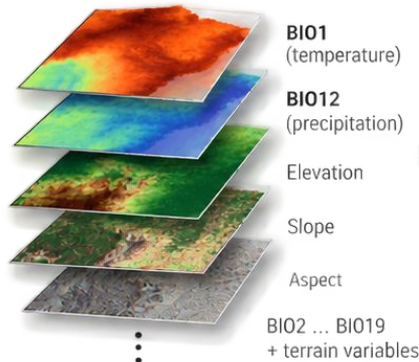


Classification: Araucaria presence



Predictor variables

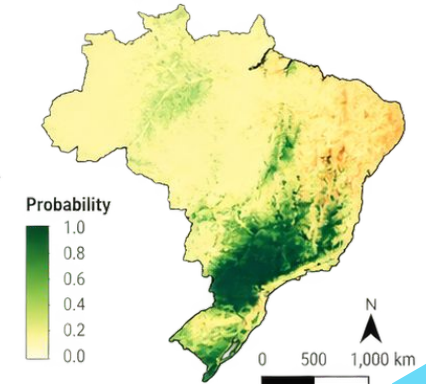
BIO1 - BIO19 + terrain variables



Machine learning classification model

Learn the relationship between environmental predictors and Araucaria occurrence (presence / absence-background)

Predicted probability of occurrence



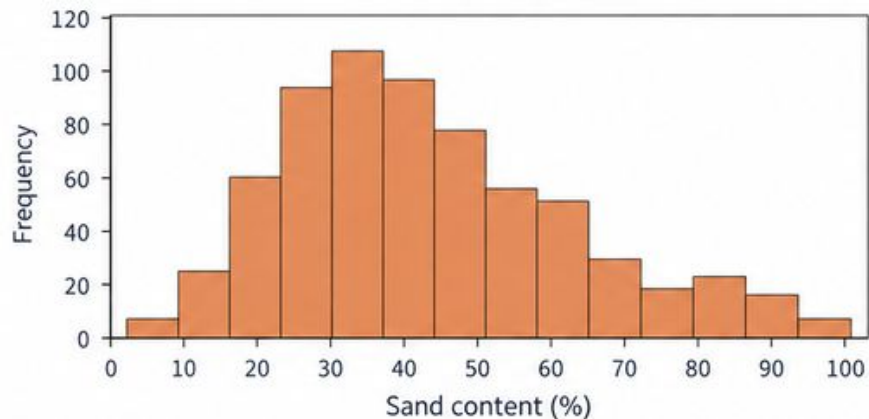
EDA: Explore Y and X_n



Explore the Response (Y)

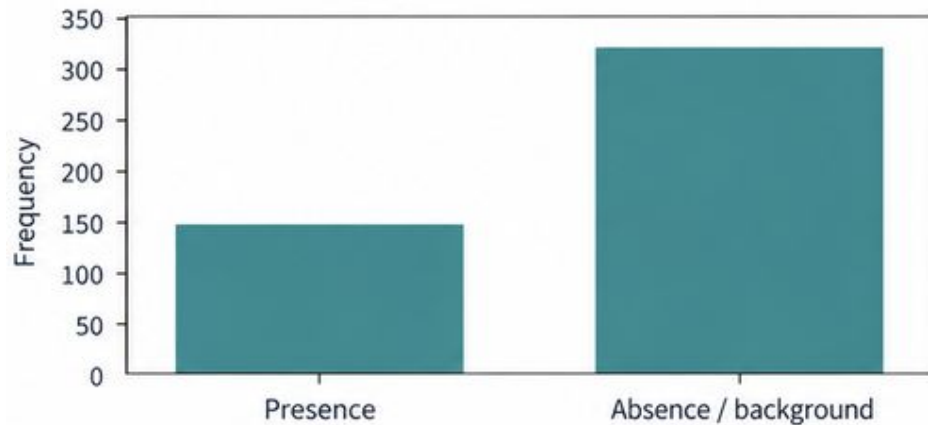
What are we trying to predict?

Regression — sand (%)



- Distribution
- Range and extreme values
- Missing values

Classification — Araucaria



- Class frequencies
- Class imbalance
- Missing values

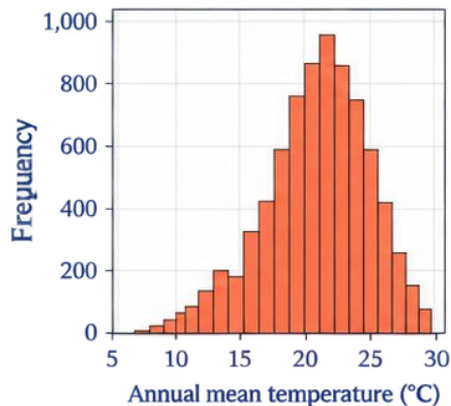
Explore the Predictors (X_n)

What information is available to explain Y ?

Distributions and ranges

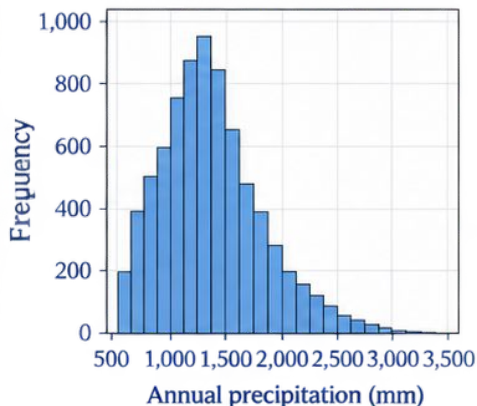
BIO1

Annual mean temperature

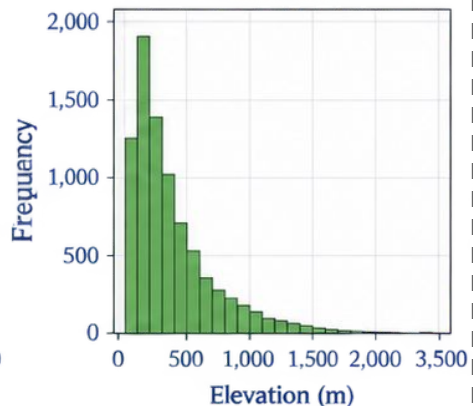


BIO12

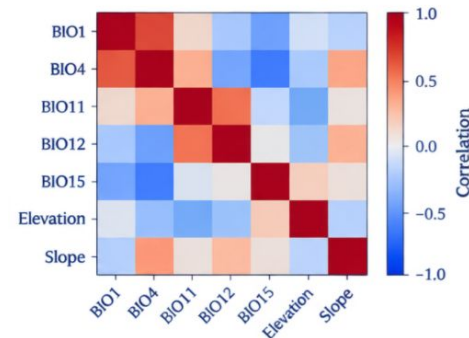
Annual precipitation



Elevation



Redundant / correlated predictors



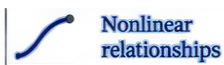
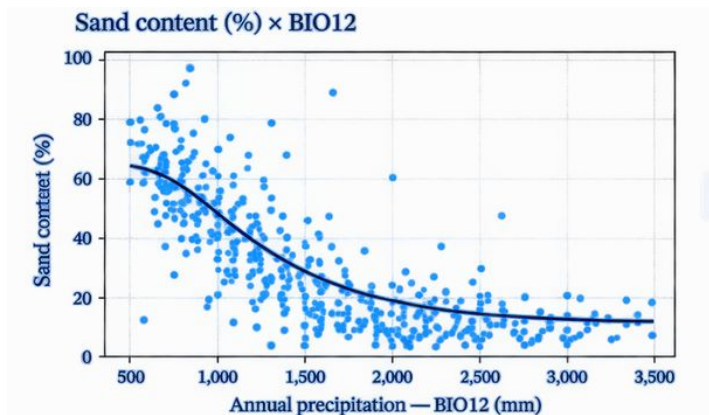
Missing values

BIO1	BIO4	BIO11	BIO12	BIO15	BIO17	BIO19	Elevation	Slope	Aspect	Curvature	HAND	Dist. to river	Dist. to urban
0.0%	0.0%	0.0%	0.0%	0.3%	0.0%	0.0%	1.8%	2.1%	0.5%	0.5%	0.0%	0.0%	0.7%

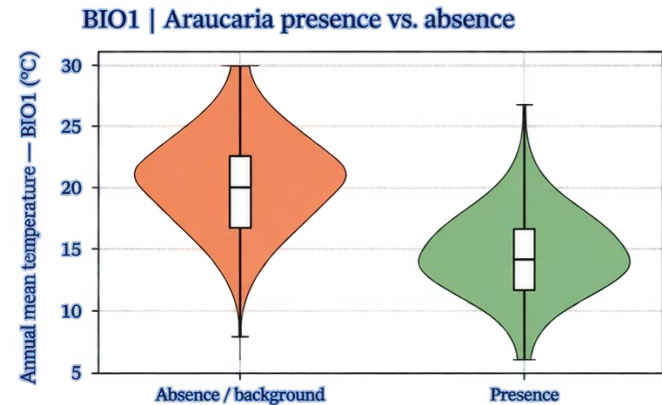
Explore relationship between X_n and Y

How does Y vary with the predictors?

Regression



Classification



Predictor distributions
by class

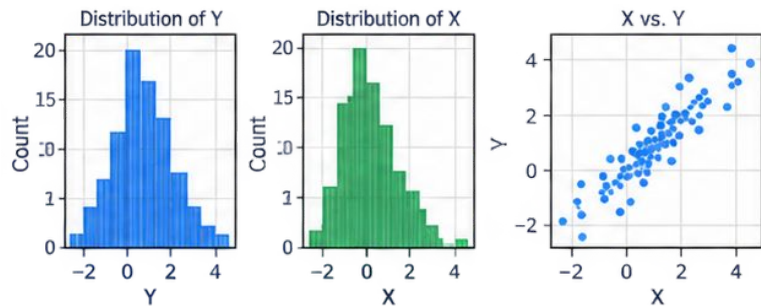


Class separation /
overlap

From EDA to ESDA

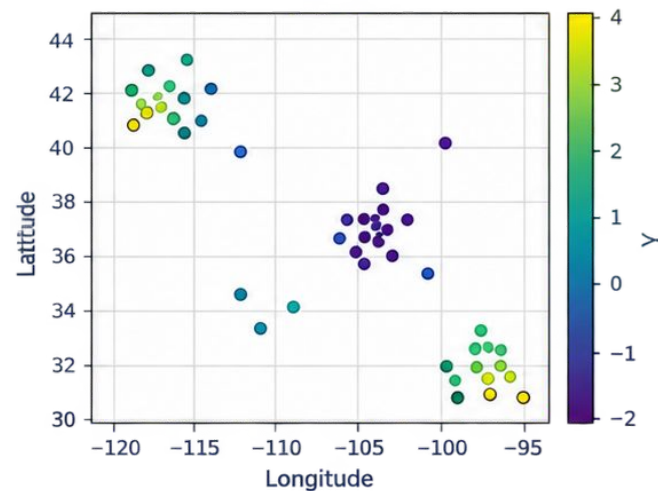
EDA

$$X_n \leftrightarrow Y$$



$(X, Y) + \text{WHERE?}$

ESDA

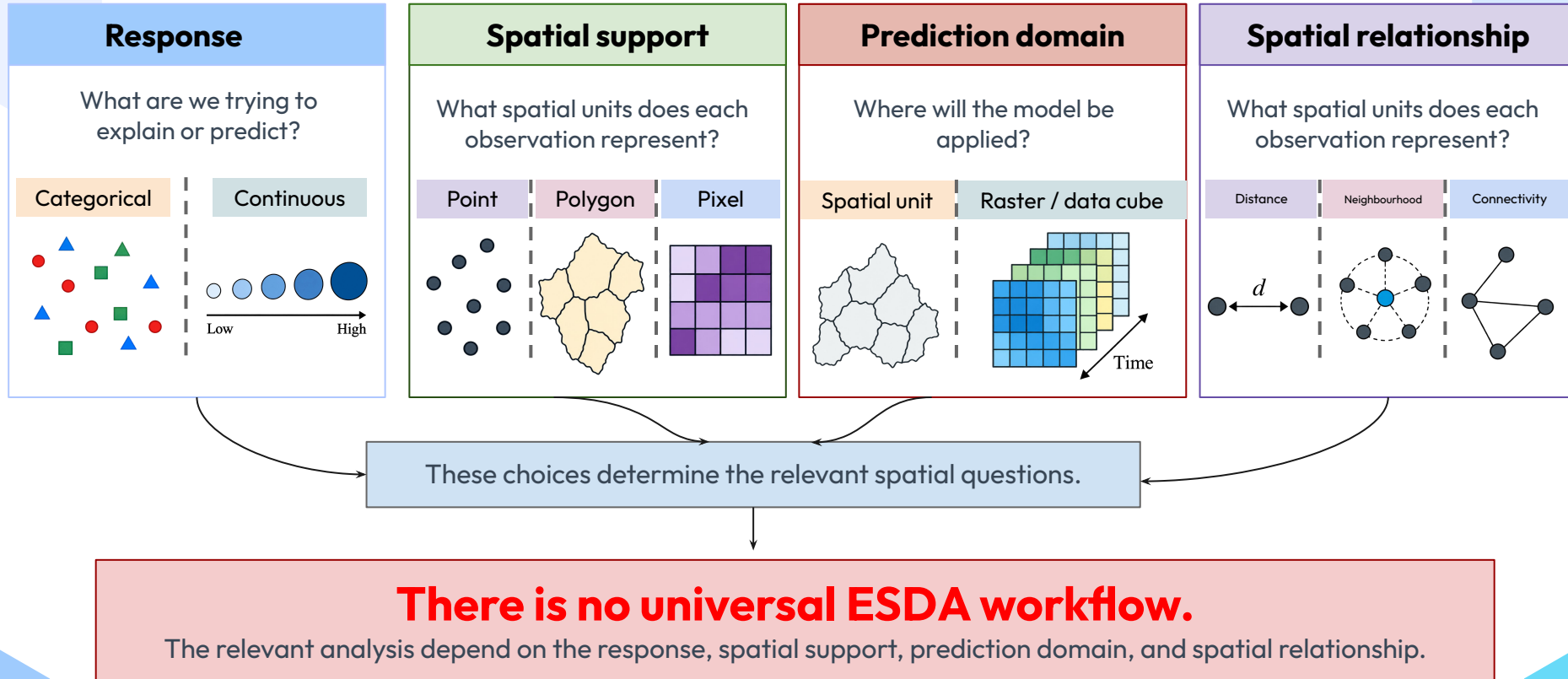


What changes when we add “Where?”



**ESDA depends
on the problem**

ESDA depends on the problem



Points patterns



Point data

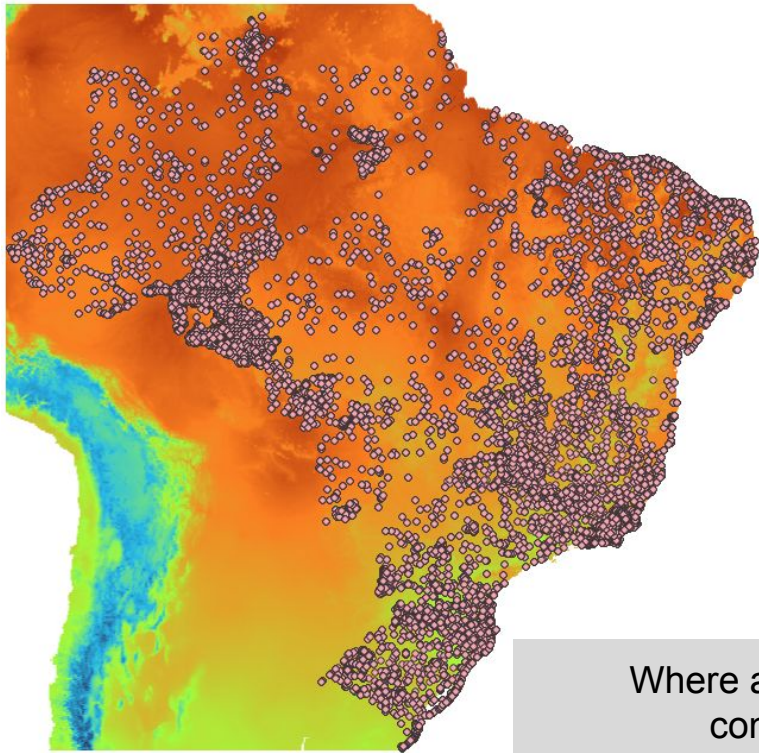
Spatial distribution of the observation not the value!

Location where disease emerged; how trees are distributed in space.

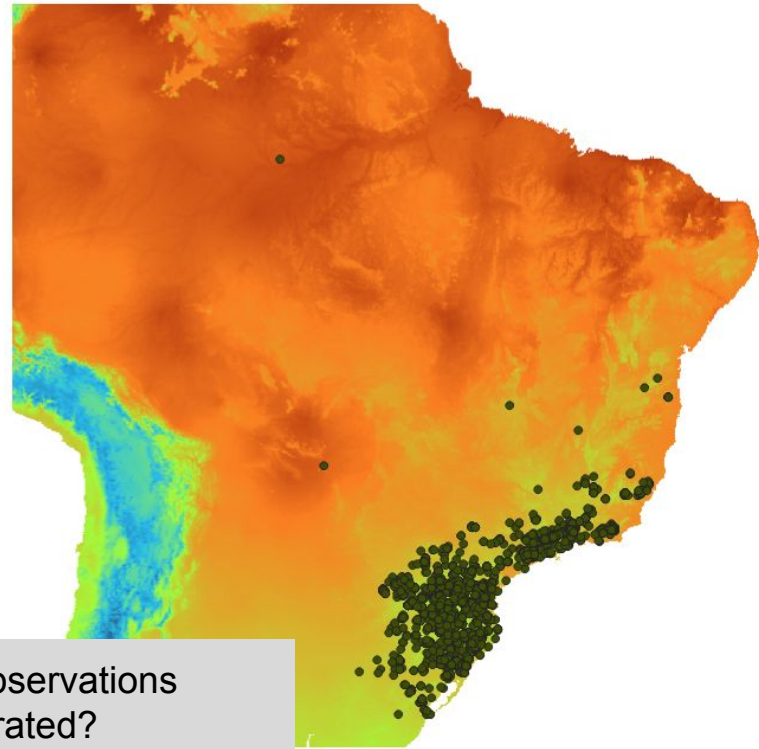
"Quantifying the gap between the spatial distribution and completely random distribution in space"

Where are the observations?

Sand samples



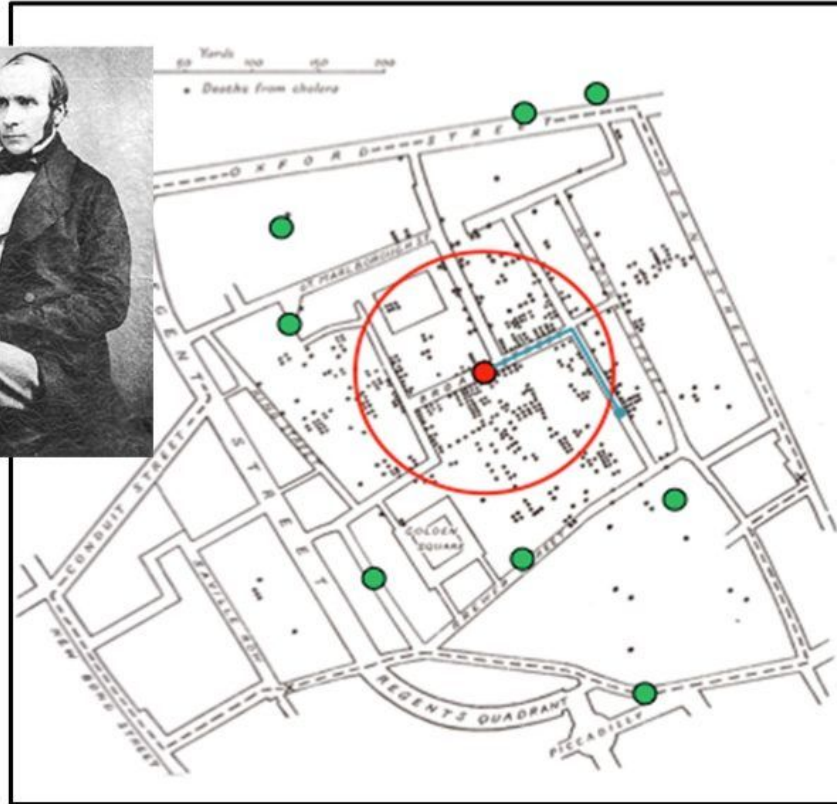
Araucaria occurrences



Where are observations
concentrated?
Where are the spatial gaps?

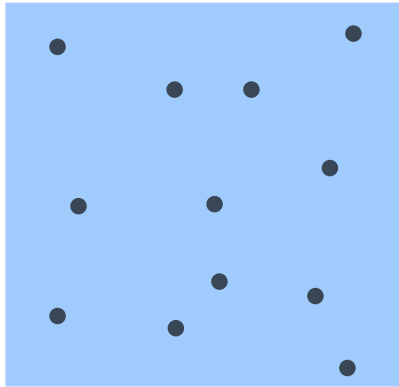
This historical map illustrates the Regent's Quadrant in London during the 19th century. The map shows a network of streets including Oxford Street, Regent Street, Piccadilly, and the Regent's Quadrant itself. Key locations marked include St. Mark's Church, Golden Square, and various streets such as Conduit Street, Saville Row, and New Bond Street. The map is overlaid with symbols indicating the locations of cholera pumps (marked with 'X') and deaths from cholera (marked with dots). A scale bar at the top indicates distances in yards, ranging from 0 to 200. The map also shows the locations of several churches and public buildings, as well as the Regent's Canal and the Regent's Park.

Point data

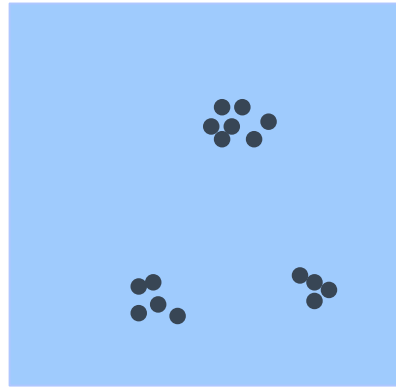


Point patterns/distribution

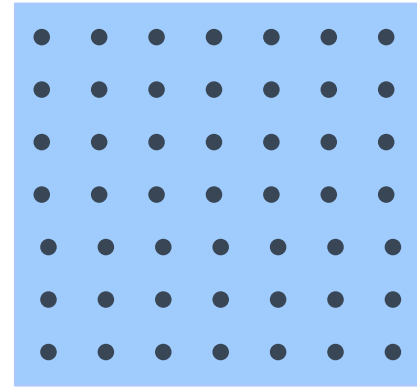
Random



Clustered

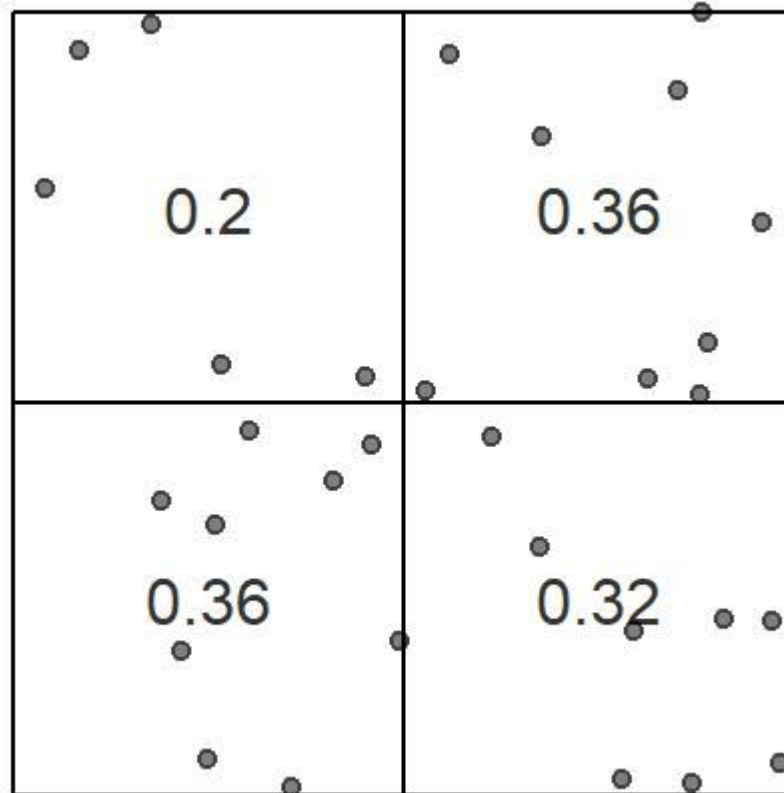


Regular

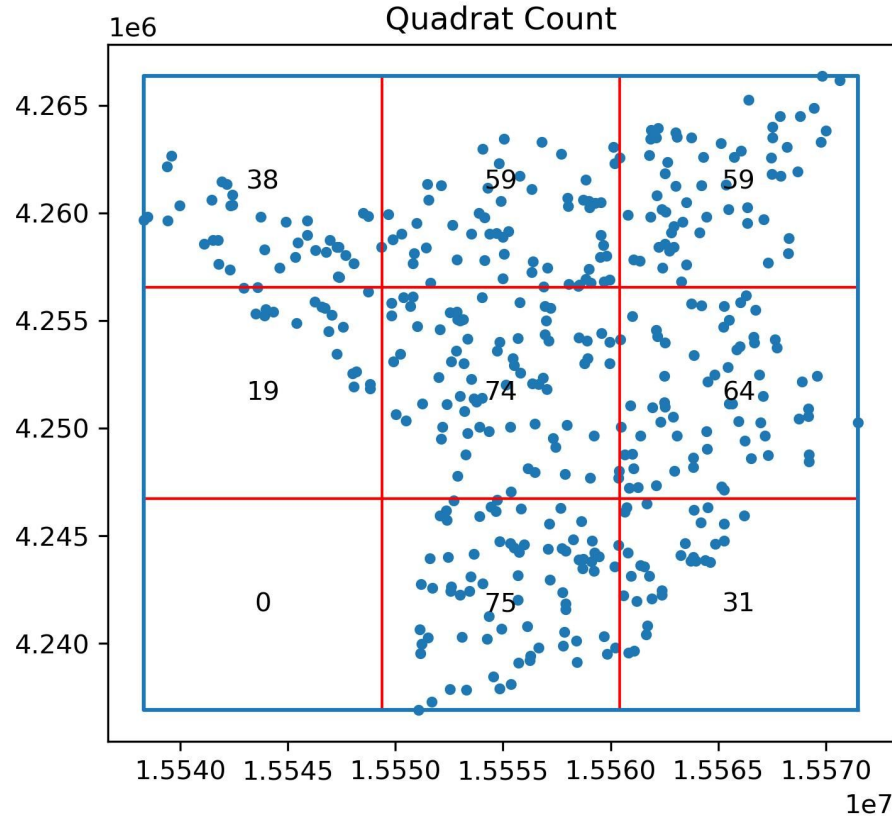


This describes the locations, not necessarily the attribute values.

Point data: quadrat analysis

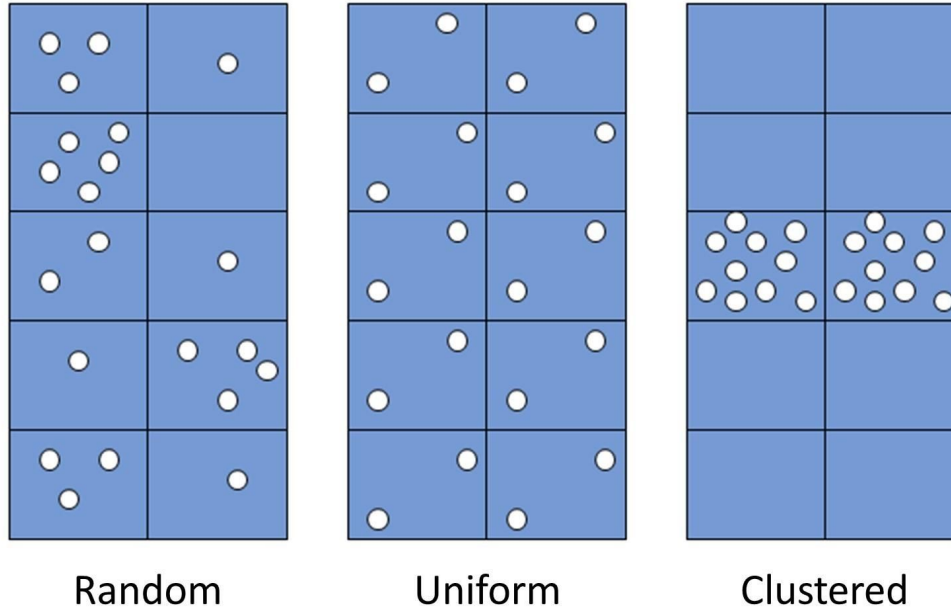


Point data: quadrat analysis



Point data: quadrat analysis

Point Patterns



Point data: Ripley's K

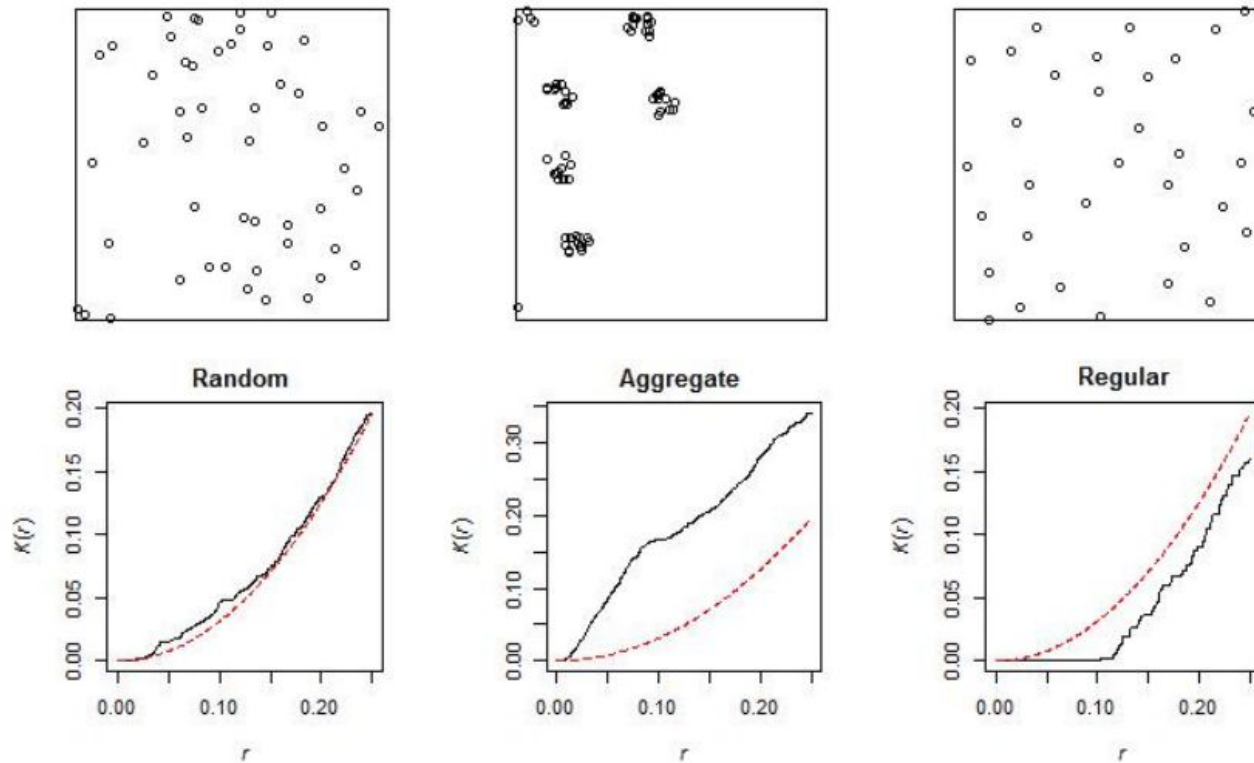
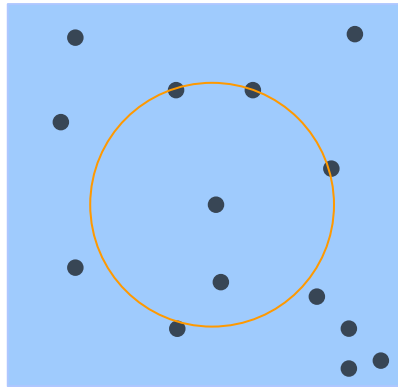
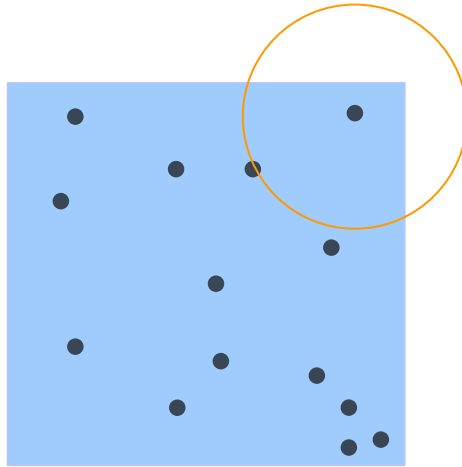


Figure 4.8 – K functions for the three standard configurations of points

Point data: Edge Effect



Point data: Edge Effect



Point data

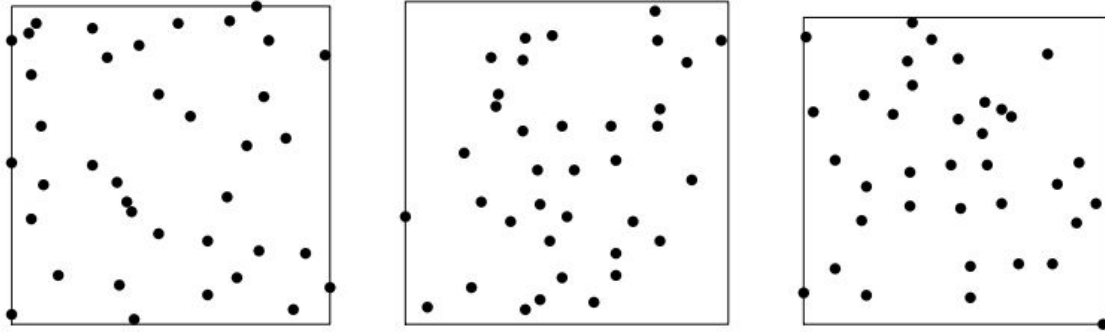


Figure 1.2. *Water strider larvae recorded from three photographs of the middle of a pond. Each frame is about 48 cm across. Data recorded by M. Nummelin, University of Helsinki, and kindly contributed by A. Penttinen.*

Point data

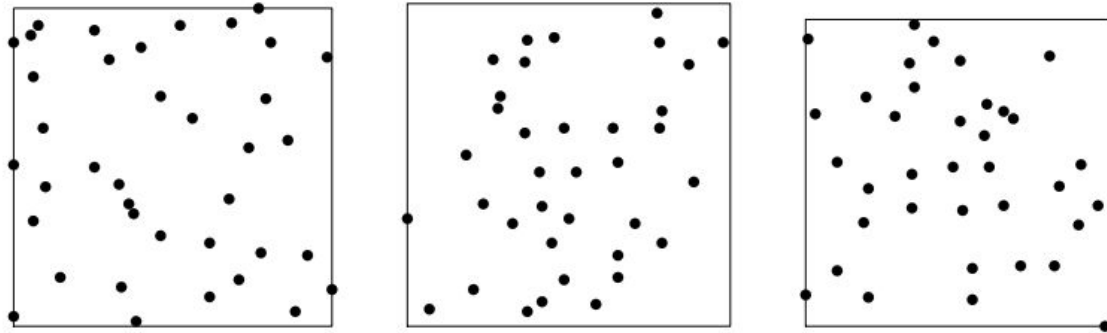


Figure 1.2. *Water strider larvae recorded from three photographs of the middle of a pond. Each frame is about 48 cm across. Data recorded by M. Nummelin, University of Helsinki, and kindly contributed by A. Penttinen.*

How to verify if the larvae are territorial?

Point data

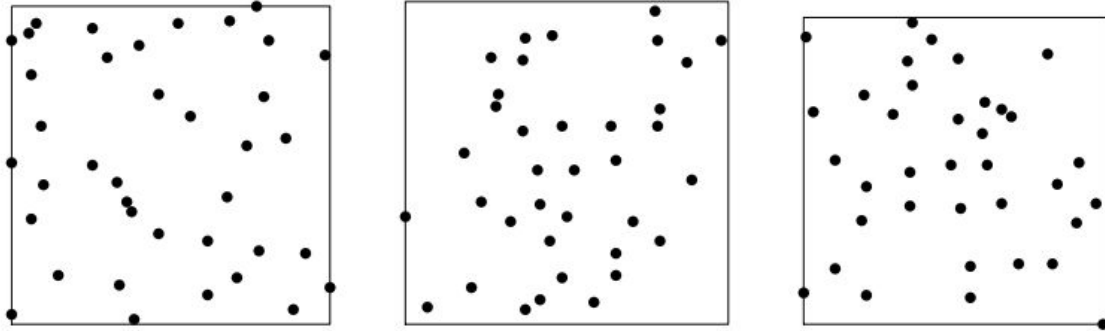


Figure 1.2. *Water strider larvae recorded from three photographs of the middle of a pond. Each frame is about 48 cm across. Data recorded by M. Nummelin, University of Helsinki, and kindly contributed by A. Penttinen.*

How to verify if the larvae are territorial?
nearest-neighbour distances are 55, 49 and 54

Point data

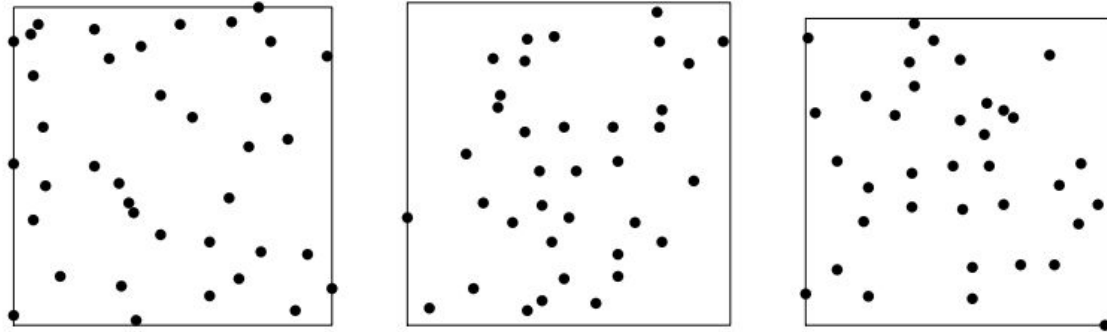


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How to verify if the larvae are territorial?

nearest-neighbour distances are 55, 49 and 54

normalise the values, dividing the observed distance by the benchmark distance for each pattern (Clark-Evans index)

Source: Baddeley, A.; Rubak, E.; Turner, R. Spatial Point Patterns: Methodology and Applications with R. CRC Press, 2015.

Point data

*****reducing a spatial point pattern to a single number, we discard a lot of information**

Point data

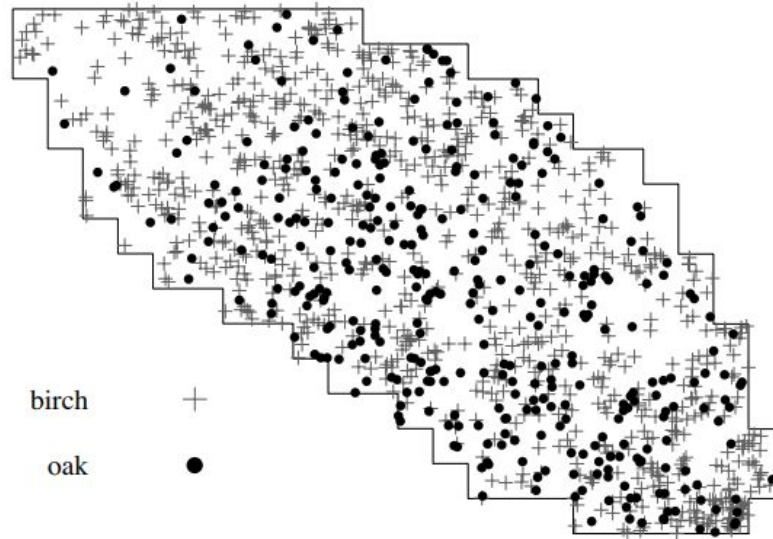


Figure 1.4. *Urkiola Woods data: locations of birch and oak trees in a secondary-growth forest in Urkiola Natural Park. Map about 220 metres across. Data collected by N.A. Laskurain, and kindly communicated by M. de la Cruz Rot.*

Point data: kernel density estimation

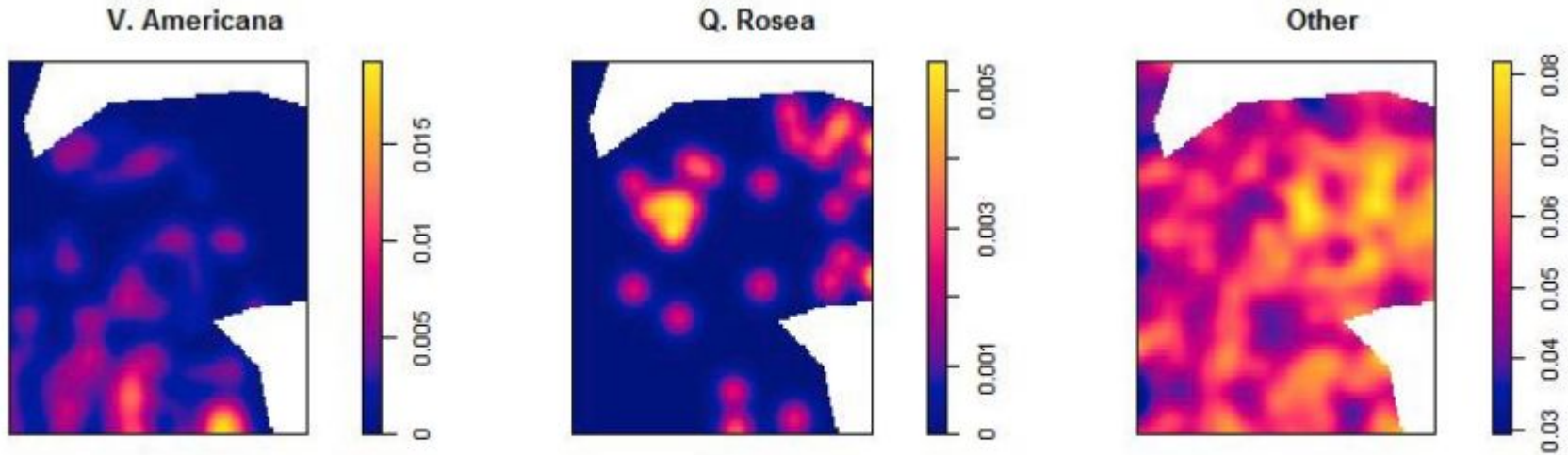


Figure 4.15 – Representation of the density of tree species *Vacapoua americana*, *Qualea rosea* or other (mix) in the Paracoul6 forest system.

Geographic Space $\neq$ Feature Space

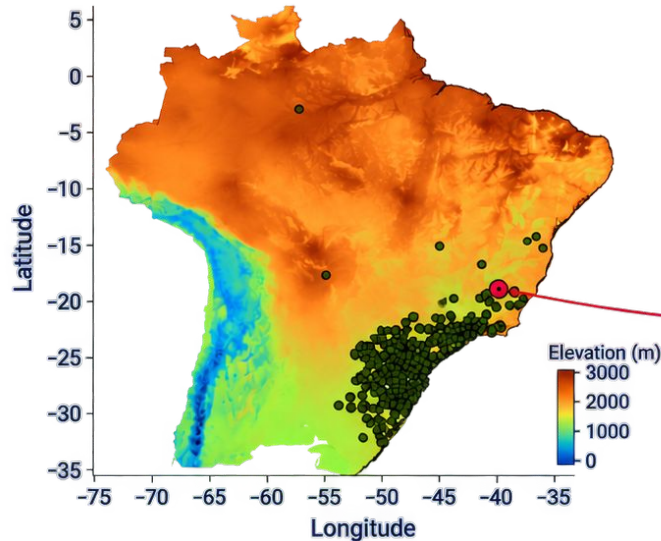


Two spaces, two questions

Where were observations collected - and which conditions do they represent?

Geographic space

Where are the observations?



Geographic
space

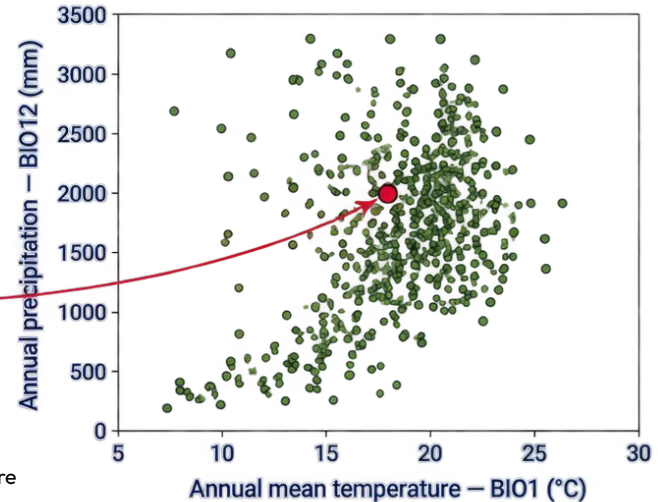
≠

Feature
space

Each observation has both a location and a position in feature space.

Feature space*

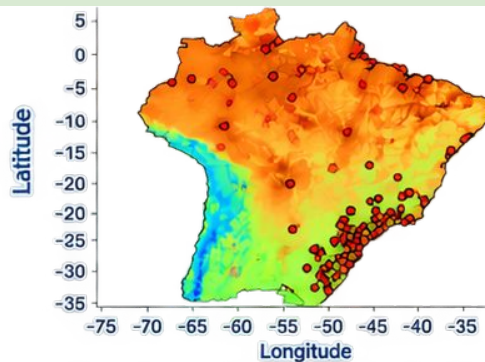
Which environmental conditions were sampled?



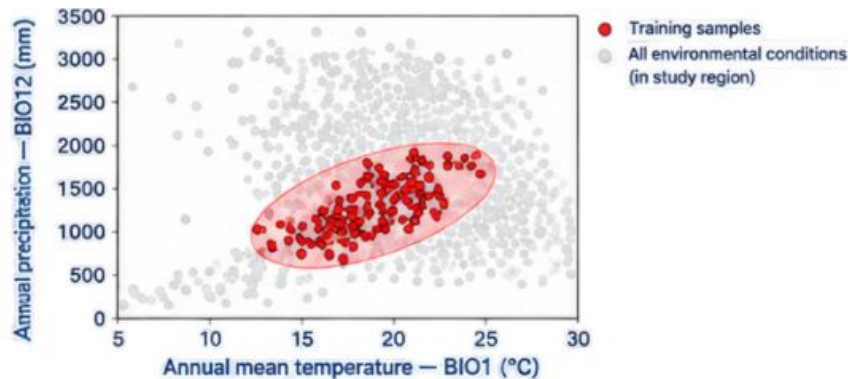
Limited geographic coverage ≠ necessarily limited feature-space coverage

Geographic coverage $\neq$ Feature-space coverage

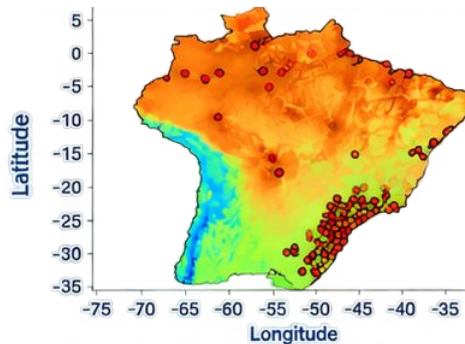
Good geographic coverage



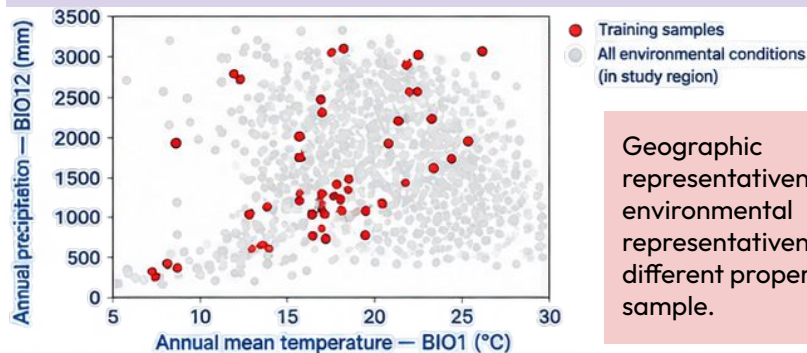
Limited feature-space coverage



Limited geographic coverage



Broader feature-space coverage



Geographic representativeness and environmental representativeness are different properties of a sample.

Prediction difficulty depends on similarity to the training data — geographically, environmentally, or both.

MOVING BEYOND SPATIAL AND RANDOM CROSS-VALIDATION IN ENVIRONMENTAL MODELLING: A CALL FOR PREDICTION-DOMAIN ADAPTIVE EVALUATION

A PREPRINT

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See:

<https://doi.org/10.48550/arXiv.2605.13689>

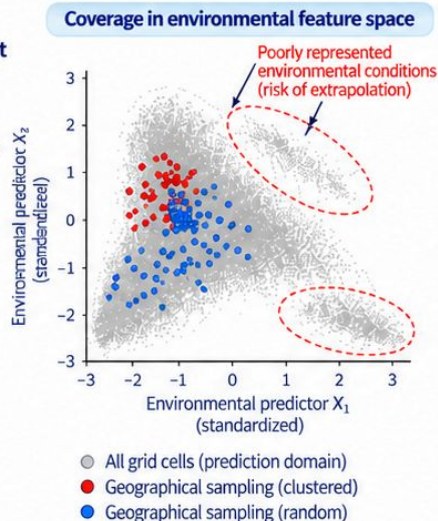
May 14, 2026

Sampling in feature space: cluster-based stratification

How to collect samples to ensure environmental representativeness and prevent feature extrapolation?

1. The Challenge: Covariate Space Bias

- Geographic random sampling or spatially clustered observations may **fail to represent important portions of the environmental (covariate) feature space**.
- Rare, transitional, or extreme combinations of predictors may remain poorly sampled.
- Consequently, the training data may cover only part of the environmental conditions occurring across the prediction domain.
- This increases the **risk of feature-space extrapolation** and may **reduce the Area of Applicability (AOA)** of the model.



Meyer et al. (2019); Ludwig et al. (2023); Milà et al. (2024).

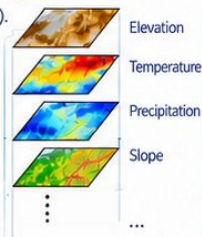
2. Cluster-based Stratification Protocol

1 Build and standardize the environmental feature space

- Stack raster covariates ($X_1, X_2, \dots, X_p$).
- Each raster cell is one observation in a p -dimensional feature space.
- Apply z-score standardization:

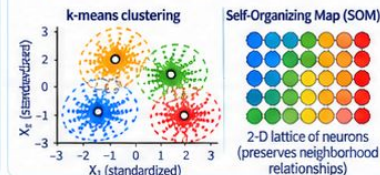
$$z = (x - \mu) / \sigma$$

- Centers each predictor around zero.
- Scales variables to comparable units.
- Prevents variables with large numerical ranges from dominating distance-based clustering.



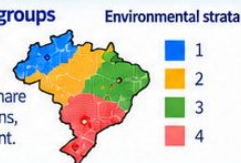
2 Unsupervised partitioning of feature space

- Cluster the standardized environmental data using unsupervised learning methods (e.g., k-means, SOM).
- Clusters are defined by covariate similarity, not geographic proximity.
- Does not require response labels (y).



3 Convert feature-space groups into geographic strata

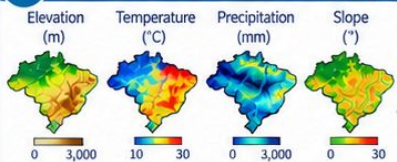
- Assign each grid cell to a cluster.
- Cells within the same stratum share similar environmental conditions, even when geographically distant.



4 Allocate geographic samples

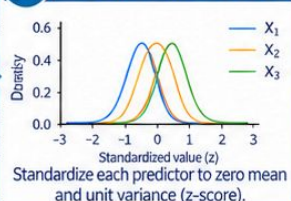
- Select field or training samples within strata.
- Strategies: equal or proportional allocation, random sampling, or centroid/medoid-based representative sampling.
- Maximize environmental representativeness across the prediction domain.

1 Environmental Rasters ($X_1 \dots X_p$)

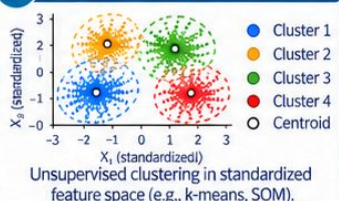


Environmental covariates X are available for every grid cell in the prediction domain.

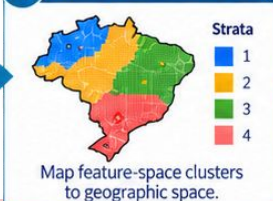
2 Z-score Standardization



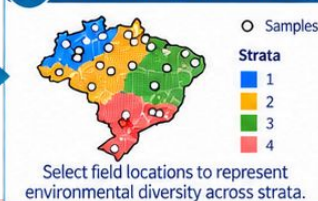
3 Unsupervised Feature-Space Clustering



4 Environmental Strata Map



5 Stratified Geographic Sampling



See: <https://dickbrus.github.io/SpatialSamplingwithR/kmeans.html>

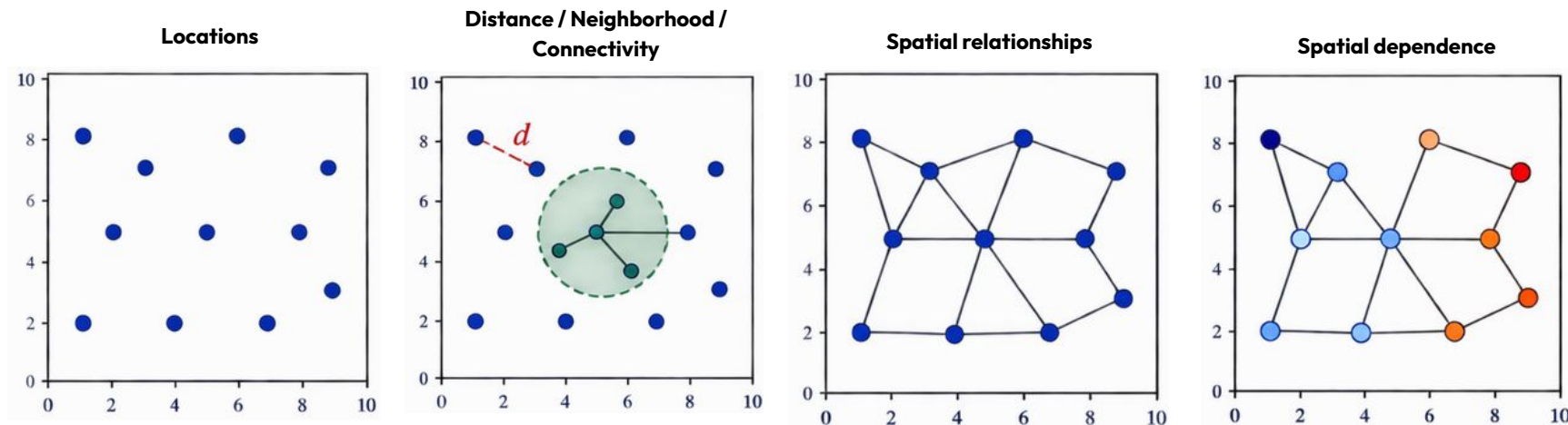
See more about: Conditioned Latin Hypercube Sampling (cLHS)

From Sampling to Spatial Structure



From sampling to spatial structure

So far we asked where observations are. Now: how are observations related in space?

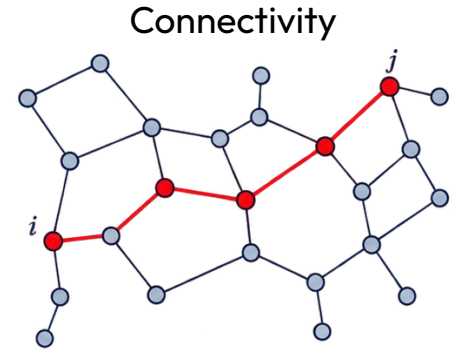
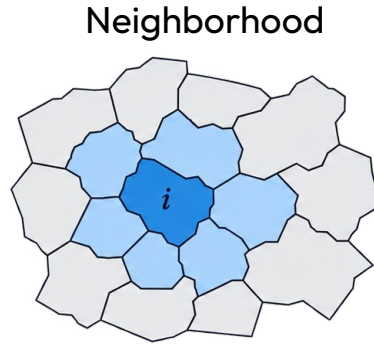
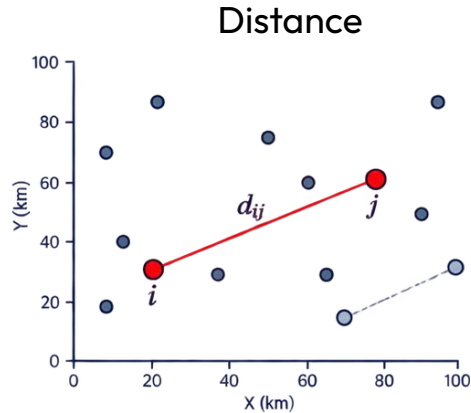


Locations tell us where observations occur. Spatial relationships allow us to ask whether their values are spatially structured.

What does “near” mean?

How should spatial proximity be represented?

Distance | Neighborhood | Connectivity



‘Near’ depends on the spatial support and the process.

From distance to neighborhood

Who is a neighbor of whom?

Distance threshold
k-nearest neighbors
Contiguity

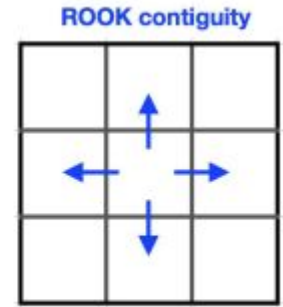
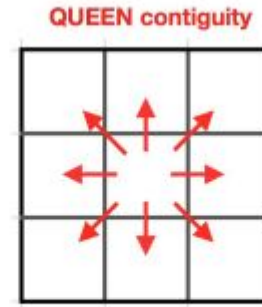
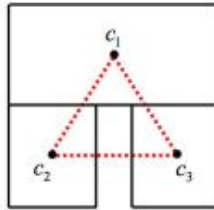
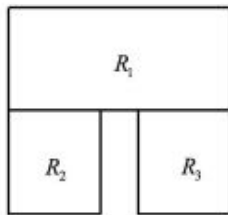


Figure 2.7 – Left: three zones - Right: distance between centroids

Source: *Smith 2016*

From distance to neighborhood

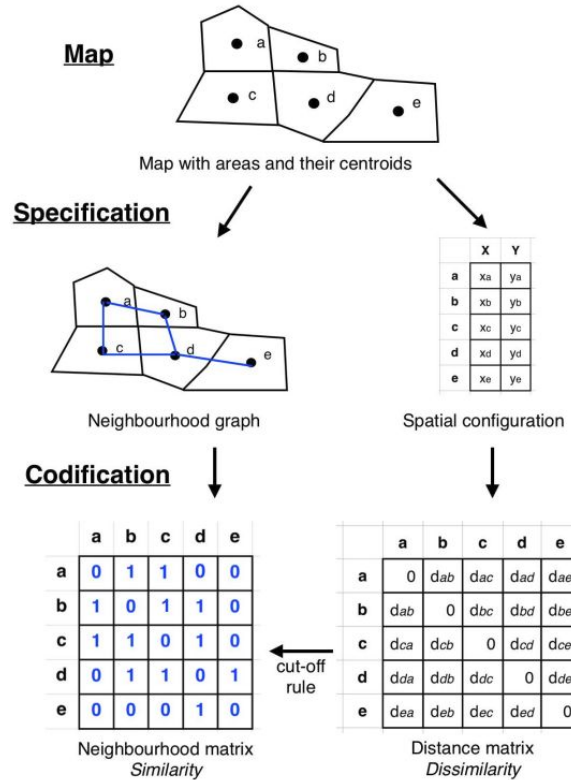


Figure 2.1 – Codifying spatial relations

Source: Tiefelsdorf 1998

From distance to neighborhood

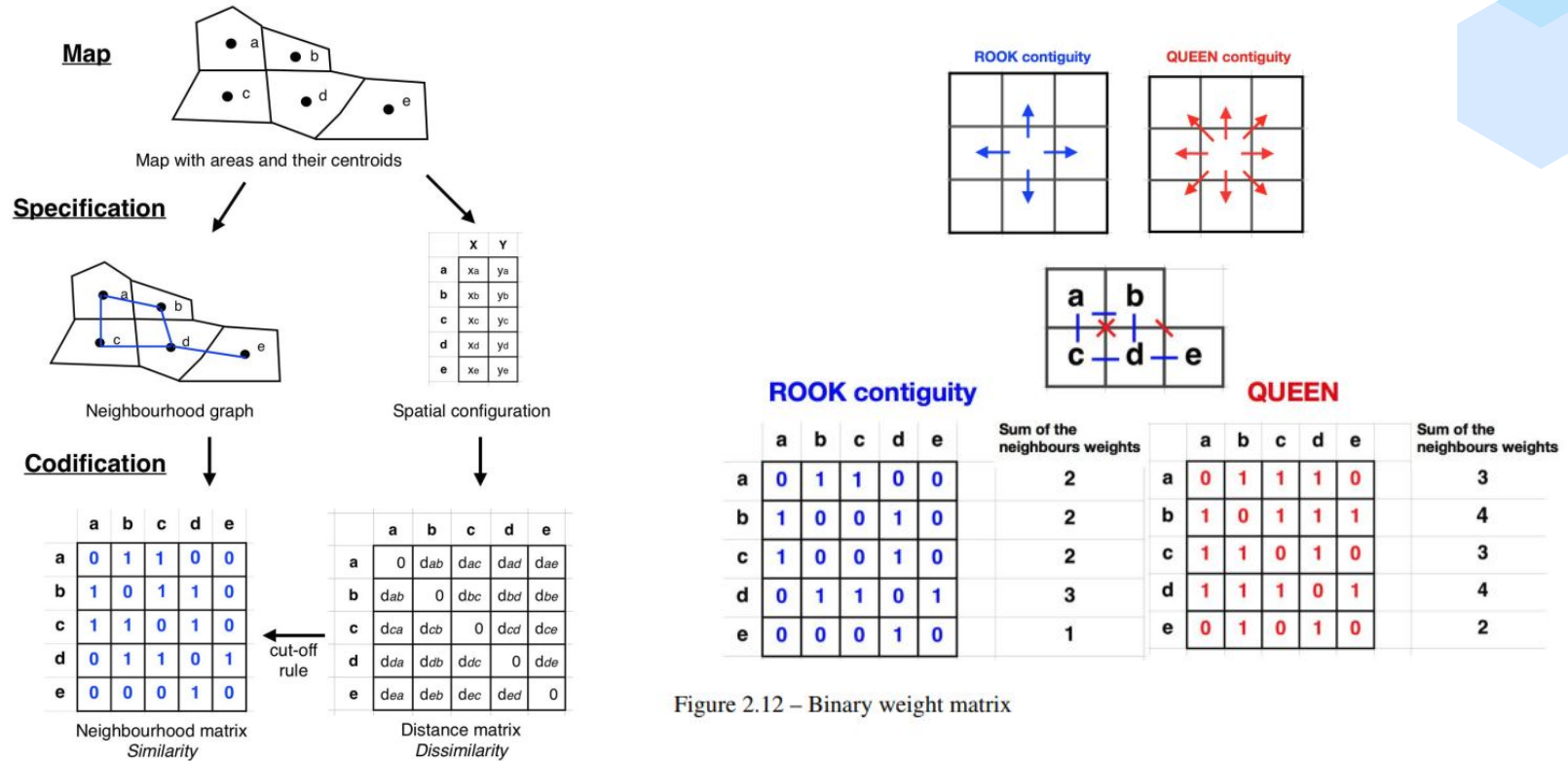


Figure 2.12 – Binary weight matrix

Figure 2.1 – Codifying spatial relations

Source: Tiefelsdorf 1998

Spatial weights

Spatial weights encode relationships among observations

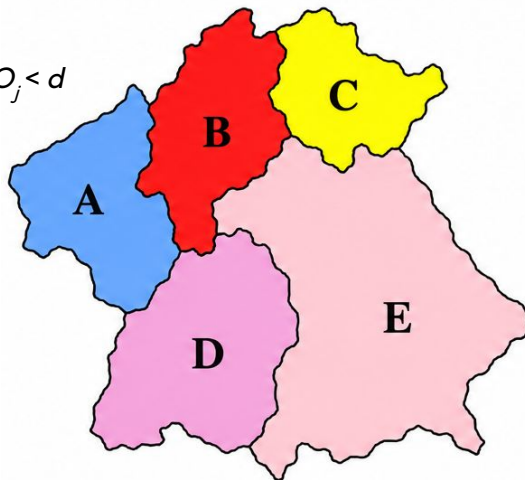
The W matrix

It has dimensions $n \times n$, where n is the number of areas (objects). The elements w_{ij} represent a measure of proximity between objects O_i and O_j .

Construction criteria:

$w_{ij}=1$, if O_i touches O_j

$w_{ij}=1$, if the distance between O_i and $O_j < d$

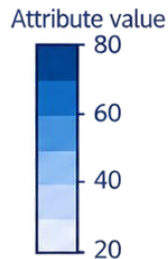
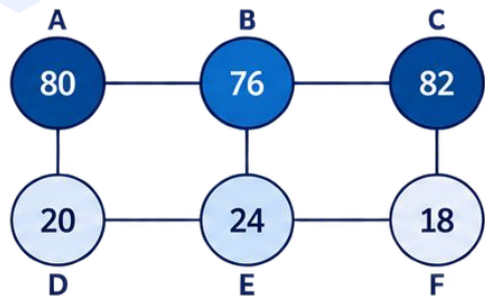


	A	B	C	D	E
A	0	1	0	1	0
B	1	0	1	1	1
C	0	1	0	0	1
D	1	1	0	0	1
E	0	1	1	1	0

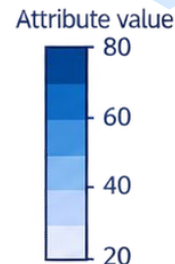
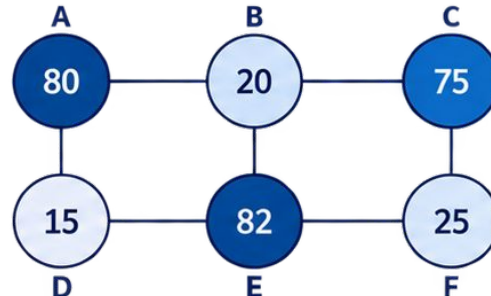
From spatial relationships to spatial dependence

Given the same relationship, are neighboring values similar?

Pattern A



Pattern B



Spatial relationship (W)
 W tells us who is related to whom

+

Attribute values (Y)
 Y tells us what values those observations have

=

Spatial dependence
Ask whether those values are systematically related through W .

How we can quantify this pattern? →



Spatial autocorrelation

Spatial autocorrelation

"absence of spatial randomness"

distribution of income, or poverty, over space

distribution of supermarkets of different brands

Spatial autocorrelation

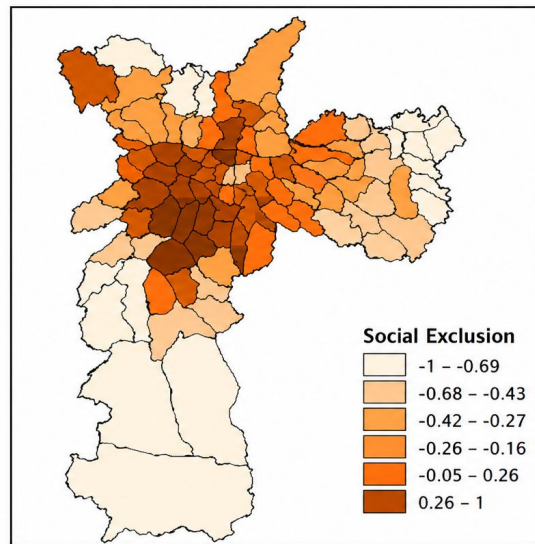
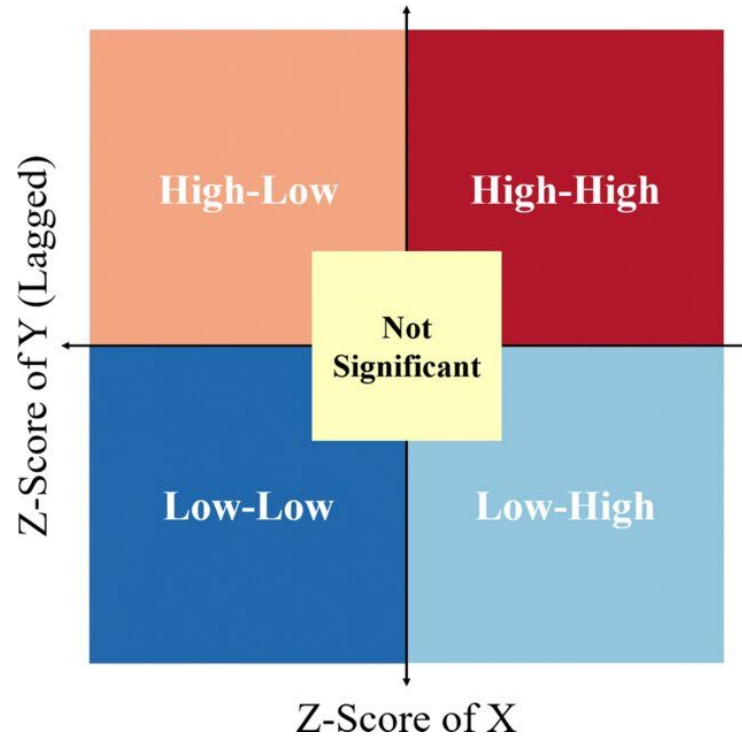


Figure 5-1 – Social Exclusion/Inclusion Index of the City of São Paulo for the districts of 1991, with 96 districts grouped by six clusters.

Moran's diagram



Moran's diagram

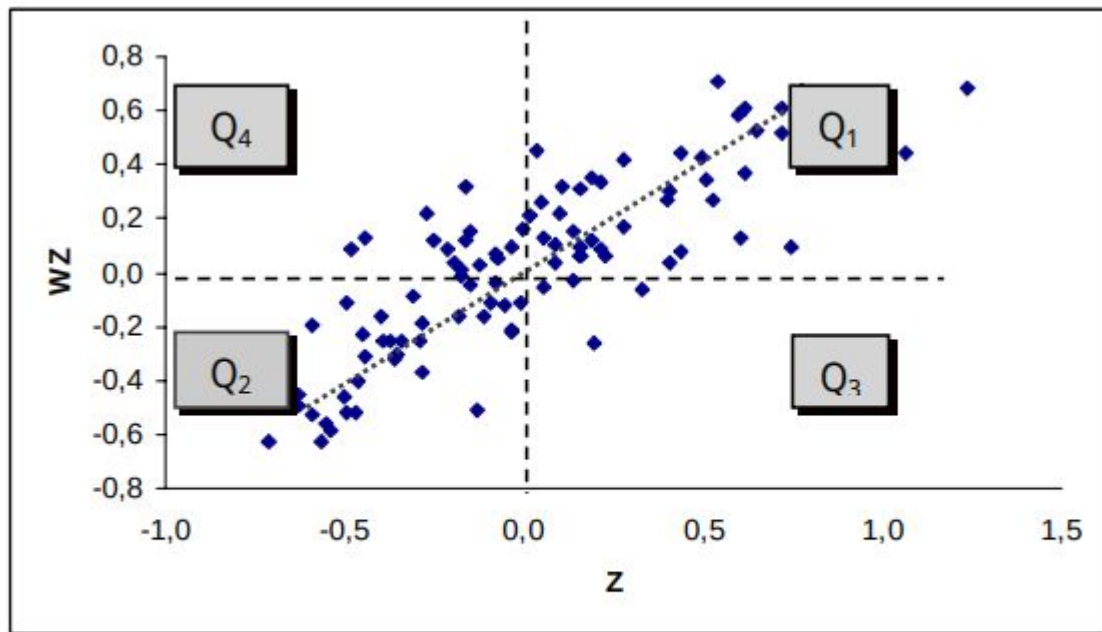
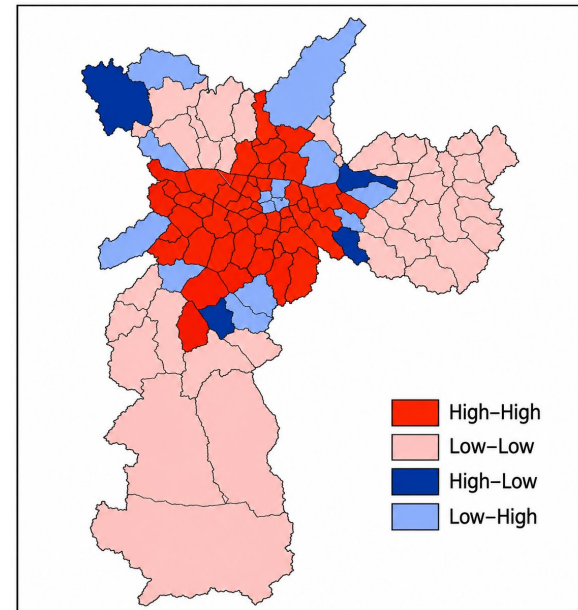


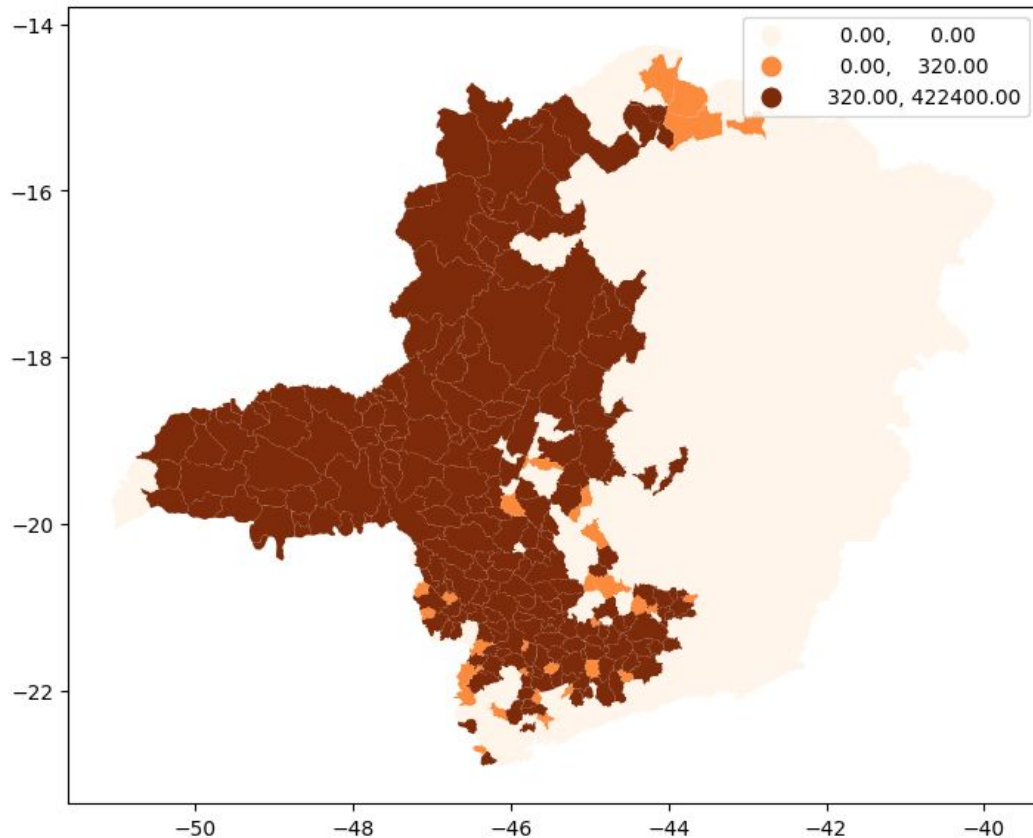
Figura 5-15 Diagrama de Espalhamento de Moran para o índice de exclusão/inclusão social de São Paulo, censo de 1991.

Figure 5-16 – Moran's Scatterplot Map for the Social Exclusion/Inclusion Index



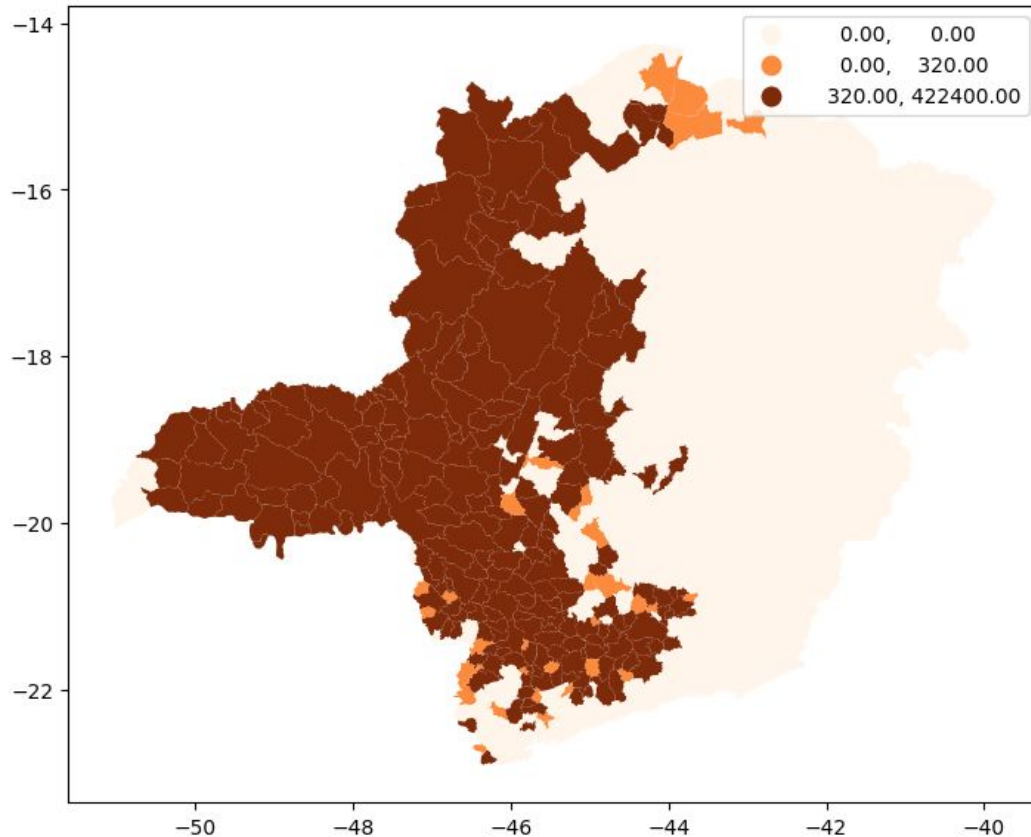
Social Exclusion Index of the City of São Paulo, 1991 Census

Spatial autocorrelation



source: Patricia Ramos (2021): https://colab.research.google.com/drive/1T8llzuL_AlxHWEs1quwH1qQg9jdQLKse?authuser=1#scrollTo=4wm22w1UKQYK

Spatial autocorrelation



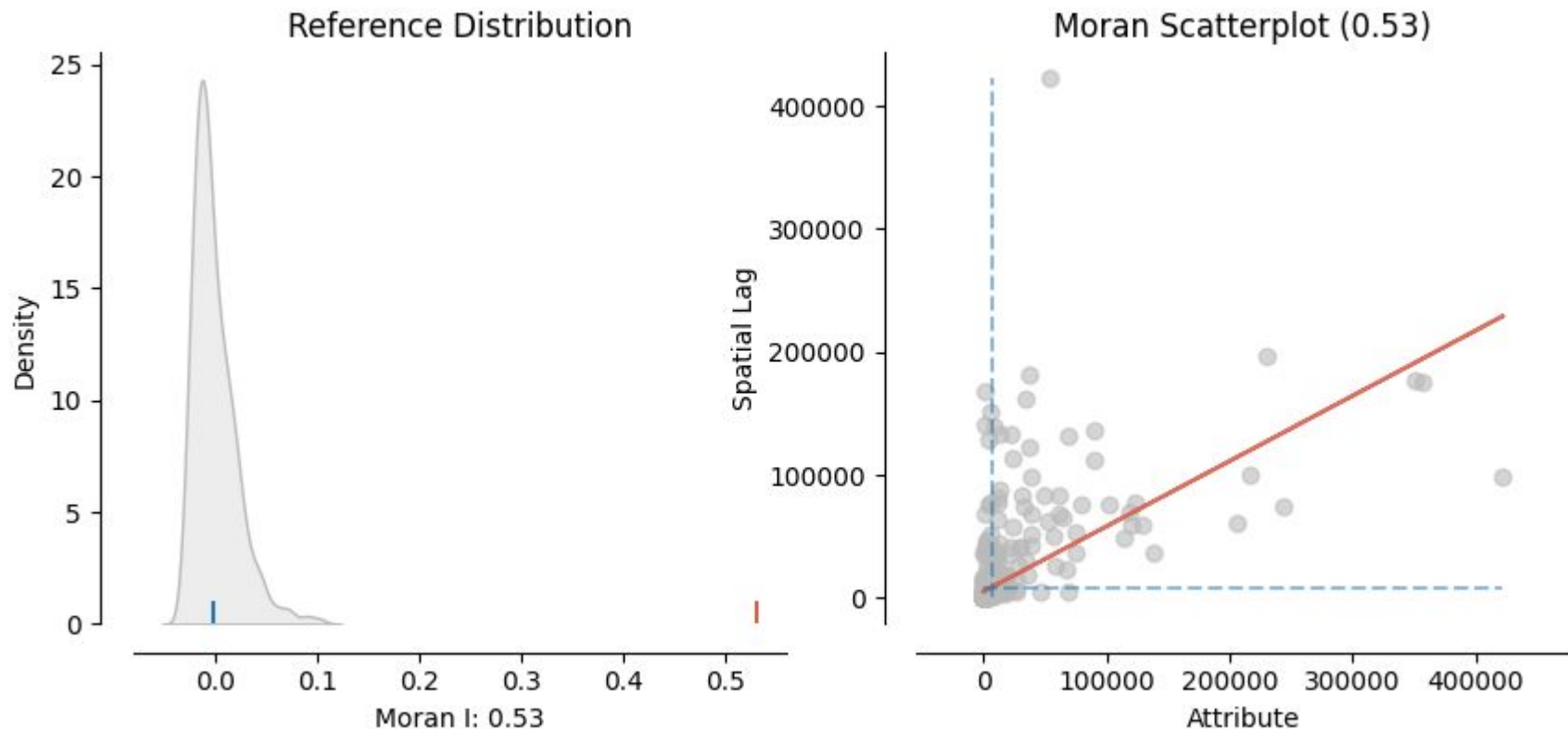
```
1 y = df[variavel].values
2 moran = Moran(y, w)
3 moran.I

np.float64(0.5304483106580684)

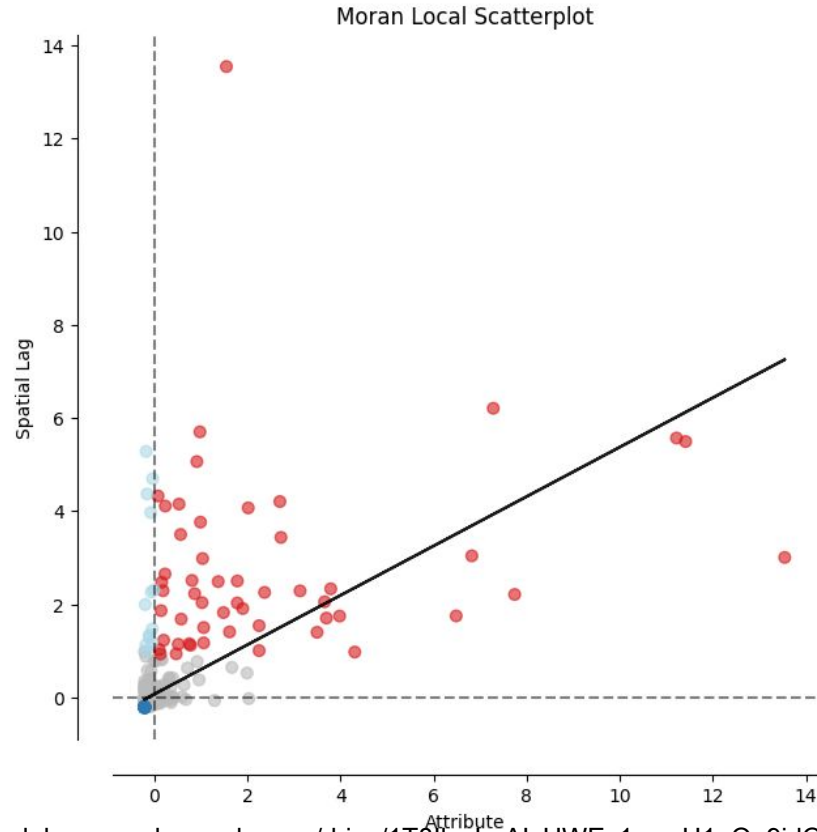
1 moran.p_sim

np.float64(0.001)
```

Spatial autocorrelation



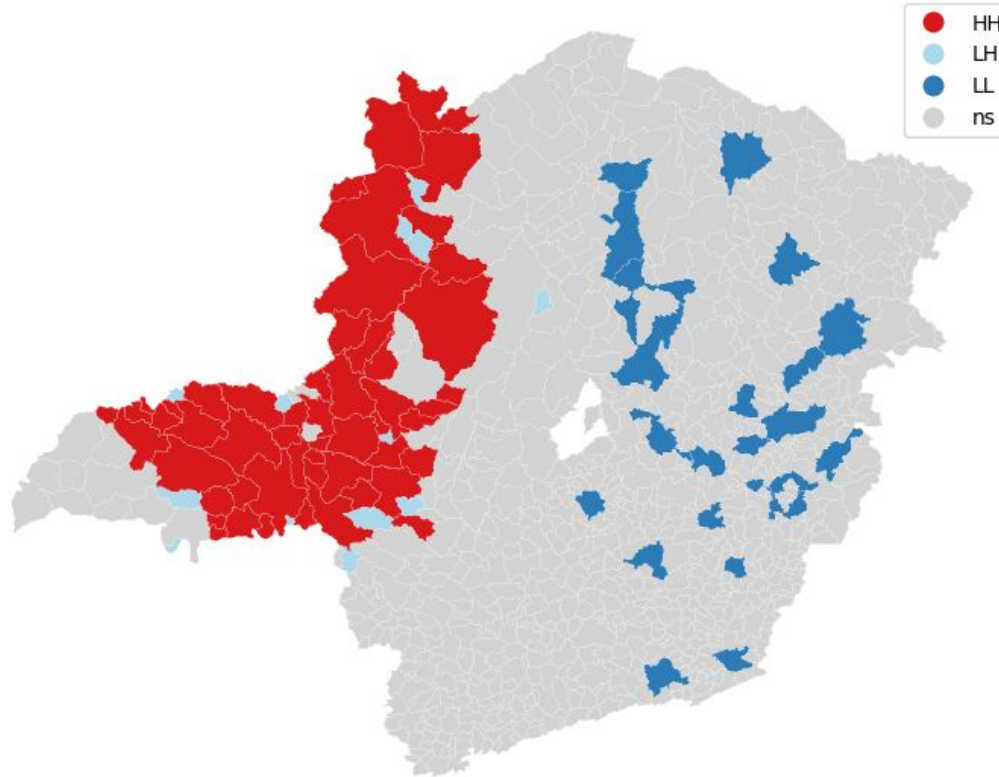
Spatial autocorrelation



source: Patricia Ramos (2021): https://colab.research.google.com/drive/1T8IzuL_AIxHWEs1quwH1qQg9jdQLKse?authuser=1#scrollTo=4wm22w1UKQYK

Local Moran/LISA

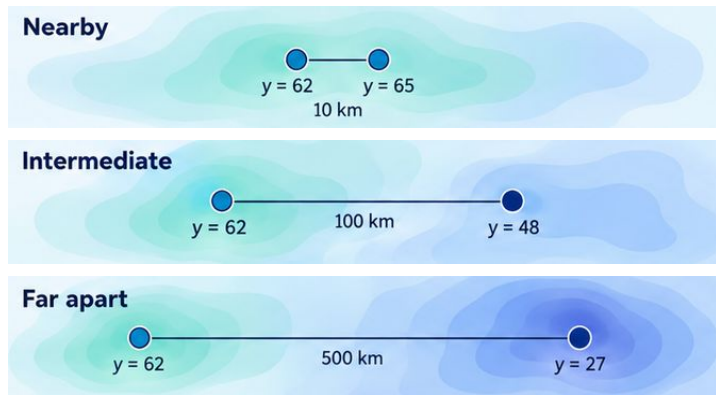
Local Indicators of Spatial Association





**Does spatial
dependence change
with distance?**

Does spatial dependence change with distance?



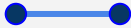
Are observations that are closer together more similar than observations farther apart?

Distance increases $\rightarrow$ similarity may decrease

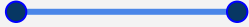
Moran's Index allowed us to ask whether neighboring observations tend to have similar values. But what if we want to know how this relationship changes continuously with distance?

Semivariogram

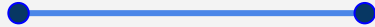
1



Short distance



Intermediate distance



Long distance

$distance = h$
 $Y(s_i) - Y(s_j)$

2

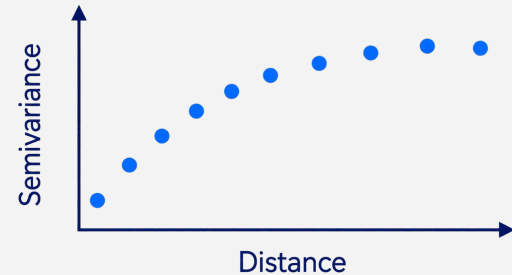
Distance (h)	Pairs
0–50 km	184
50–100 km	267
100–150 km	321
...	...

We compare many pairs of observations separated by approximately the same distance.

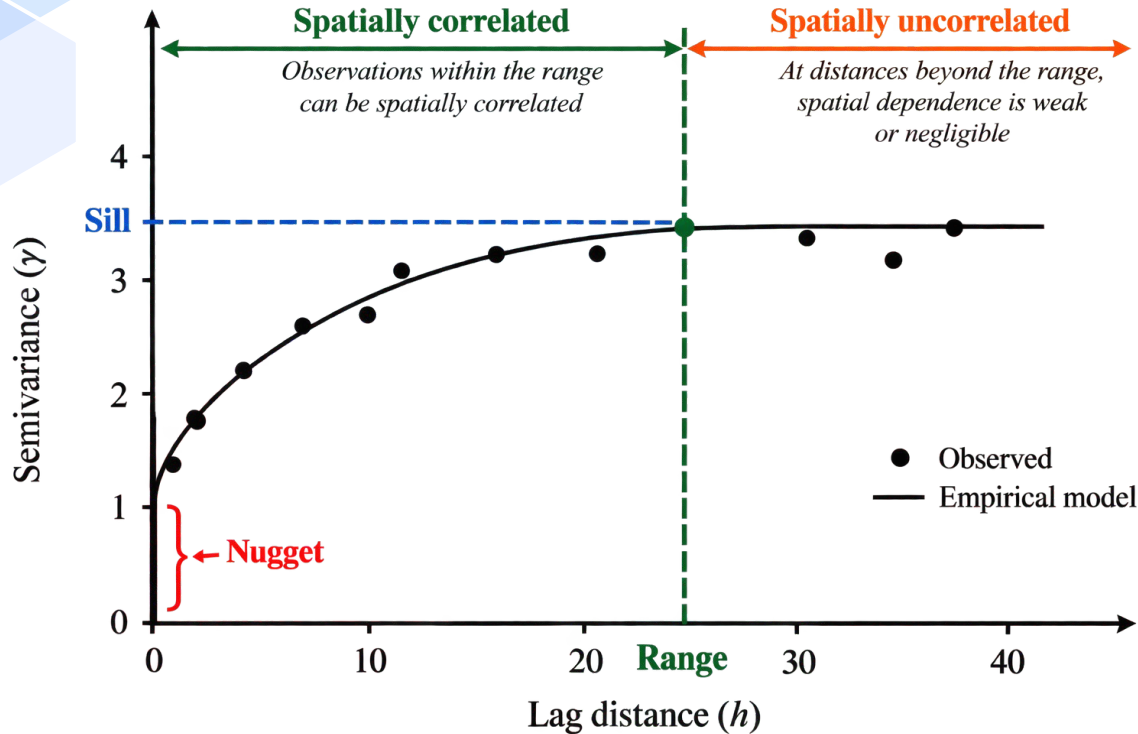
3

$$\hat{\gamma}(h) = \frac{1}{2N(h)} \sum_{i=1}^{N(h)} [Z(x_i) - Z(x_i + h)]^2$$

Semivariance = average dissimilarity between observations separated by distance h .



Semivariogram



Semivariogram Range → Modeling Decisions

- **Spatial scale:** approximate distance over which spatial dependence persists.
- **Neighborhood definition:** informs meaningful search distances or spatial context.
- **Sampling design:** helps assess whether sampling spacing captures spatial variation.
- **Validation:** can inform spatial separation between training and test data.
- **Spatial Feature engineering:** sets the window radius (kernel/neighborhood) to extract local contextual covariates.

Range helps summarize the spatial scale at which dependence may operate.

What exactly are we testing?

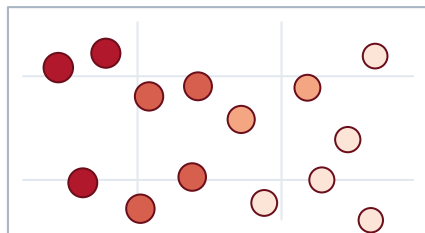
What variable are we examining for spatial structure?

"Is there spatial autocorrelation?" is incomplete without asking: spatial autocorrelation of what?



Response Y

Is the phenomenon spatially structured?

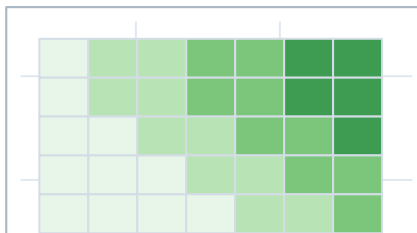


nearby observations
tend to share values

OBSERVED

Predictors X

Are explanatory variables spatially structured?

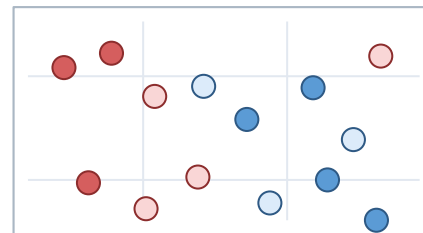


$X_1, X_2, \dots, X_n$
spatially structured
explanatory information

EXPLANATORY

Residuals / errors

Does unexplained spatial structure remain?



$e_i = y_i - \hat{y}_i$
spatial structure may
remain unexplained

UNEXPLAINED

Spatial autocorrelation has a different interpretation depending on what is being analyzed.

Y describes the phenomenon, X describes the explanatory information, and residuals describe what the model did not explain.

WHAT IS SPATIALLY STRUCTURED?

Y Phenomenon

≠

X Predictors

≠

e Unexplained variation

Where does Moran's Index fit?

Geospatial ML problem

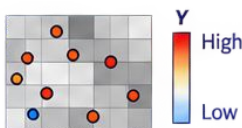
Polygon regression (continuous Y)



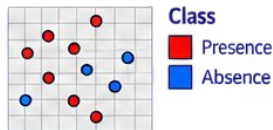
Polygon classification (categorical Y)



Point → raster regression (sample points)



Point → raster classification (sample points)



Before modeling

Moran's I of Y / relevant X

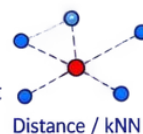
- Does the continuous response exhibit spatial structure?
- Are relevant explanatory variables spatially structured?

Spatial association of classes / X

- Are classes spatially clustered?
- Are relevant explanatory variables spatially structured?

Moran's I of sampled Y using point neighbors

- Do the sampled observations exhibit spatial structure?



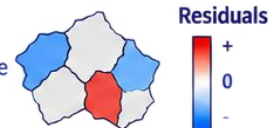
Spatial structure / scale of X

- Do predictors exhibit spatial structure?
- At what spatial scales are they structured?

After modeling

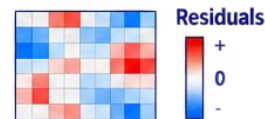
Moran's I of residuals

- Does unexplained spatial structure remain after modeling?



Spatial structure of errors

- Do prediction errors exhibit spatial structure?



Moran's Index has different roles depending on the geospatial ML problem and on what is being analyzed

ESDA → Modeling Decisions

What we learn during exploration should shape how we model and predict.

Sampling patterns

Clustering, gaps, uneven coverage

**Sampling bias & geographic
representativeness**

Feature-space coverage

Unsupported conditions

Extrapolation & Area of Applicability (AOA)

Spatial dependence

Dependence and its spatial scale

Independence assumptions & validation design

Spatial context

Distance, neighborhood, connectivity

**Spatial feature engineering or model
structure**

Residual structure

Spatial pattern in model errors

Missing spatial structure?

WHERE changes not only how we explore the data, it changes what we can expect from the model.

Exploratory (Spatial) Data Analysis

From attributes to spatial structure

Édipo Cremon
Rennan Marujo

